AG

Agrawal, Gans & Goldfarb

Economists of AI · prediction machines

Contemporary · Ajay Agrawal, Joshua Gans and Avi Goldfarb, University of Toronto.

> *"AI lowers the cost of prediction."*

THE WHY THAT DROVE THE WORK

They sought a clear economic account of what AI does, concluding that it is fundamentally a drop in the cost of prediction — not of judgment.

INTELLECTUAL FOUNDATION

Economists at the Rotman School, they reframed AI through the lens of price theory: when the cost of prediction falls, demand for it and for its complements shifts.

FRAMEWORKS ANCHORED IN THIS COURSE

They anchor the Prediction Machines Lens in Unit 6's prediction-and-risk Slibrary, and inform the prediction-judgment split across the course.

HOW THE IDEAS OPERATIONALISE

Their framing operationalises as identifying where cheaper prediction creates value and where human judgment must still be supplied.

LIMITATIONS & CRITIQUE

Treating AI purely as prediction can understate generative and agentic capabilities that blur the prediction-judgment line.

ENDURING IMPACT

Prediction Machines became a standard economic framing of AI for business and policy audiences.

ON THE SAME PANEL (UNIT 6)

- **Harry Markowitz** · Giant 24
- **Fischer Black & Robert Litterman** · Giant 25
- **Eugene Fama & Kenneth French** · Giant 26
- **Marcos López de Prado** · Giant 27

SOURCES (HARVARD REFERENCING STYLE)

Agrawal, A., Gans, J. and Goldfarb, A. (2018) Prediction Machines. Boston: HBR Press.

Agrawal, A., Gans, J. and Goldfarb, A. (2022) Power and Prediction. Boston: HBR Press.